STATISTICAL ARBITRAGE · SLEEVE 1

Mean reversion on the Treasury curve, after the hedge

A walk-forward PCA relative-value model, 1995–2026. The signal is real. The question is whether it pays for its trades.

Nile Ellis

NYU Tandon · October 21, 2026

THE ANSWER UP FRONT

The signal works before costs. It does not survive them.

Gross Sharpe, headline

0.55

Before costs. Best variant (PCA-3 sparse): 1.21

Net Sharpe, headline

-0.60

After 0.25 bp a leg. Best variant: −0.09

Breakeven cost, headline

0.48×

of the cost table, about 0.12 bp a leg. Best variant: 0.93×

Signal: −z predicts the next 10 bars with rank IC 0.17 (headline) to 0.33 (PCA-3 sparse), $p < 0.01$. **Cost:** 9.6–10.8% of capital a year, against 3.8–10.1% of gross P&L.

Out of sample, walk-forward, 1995–2026, 10% vol target

DATA AND TEST DESIGN

Thirty-one years, tested one year at a time

Data

Tenor	Data from	First traded
2y, 5y, 10y, 30y	1990	Dec 1994
3y	2005	May 2010
7y	2009	Apr 2014

Bloomberg generic Treasury yields, daily, in bp. A tenor trades only once it has a full 5-year training window. Credit: HY and IG OAS, CDX HY, HYG.

Test design

- Walk-forward: 1,260-bar train (5 years), 252-bar test (1 year), 33 folds
- 10-bar purge and 5-bar embargo between train and test
- Loadings, bands, scalers and classifier fitted on train only
- Cost: 0.25 bp per Treasury leg, one way
- Sleeve scaled to a 10% annual vol target
- Headline named in advance: PCA-2 curve plus 3 credit pairs

First test block Dec 2, 1994; last Nov 6, 2025; sample ends Sep 25, 2026

RESEARCH QUESTION

Does the residual revert enough to trade?

"After hedging level and slope with PCA, does the Treasury curve residual mean-revert enough to trade profitably after realistic costs?"

Answer: it reverts, but the reversion per trade is smaller than the round-trip cost.

0.17–0.33

Rank IC of −z with the next 10 bars on entry days (headline to PCA-3 sparse). The signal points the right way.

0.45 < 0.73

bp. PCA-2: mean realised move per trade, against the median round-trip cost.

−0.60

Headline net Sharpe at 0.25 bp a leg; 0.55 before cost.

Out of sample, walk-forward 1995–2026, 0.25 bp a leg. Sources: deck_numbers.md §E and §A (reversion vs round-trip cost).

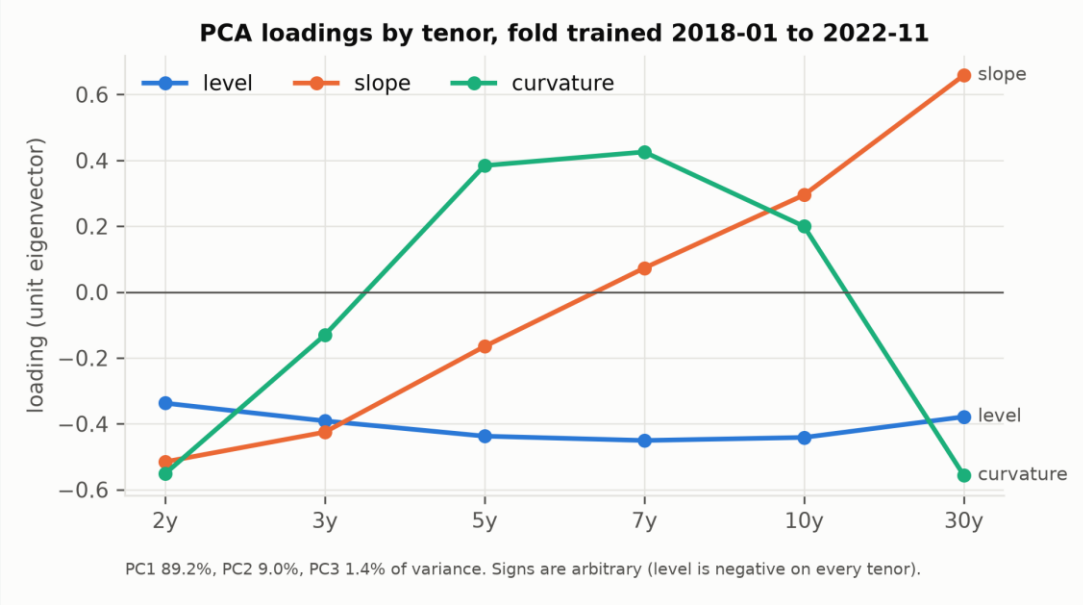
PCA · IN AND OUT

Three factors explain 99.6% of moves

Δy: daily changes, bp, 6 tenors, 1,259 days

$\Sigma = \mathrm{cov}(\Delta y) = V \Lambda V^{\top}$; L = top-k eigenvectors

Tenor	Level	Slope	Curvature
2y	−0.337	−0.516	−0.552
3y	−0.391	−0.425	−0.131
5y	−0.438	−0.165	+0.384
7y	−0.451	+0.073	+0.426
10y	−0.442	+0.295	+0.200
30y	−0.379	+0.659	−0.557
Eigenvalue, bp²	138.0	13.88	2.14
Variance share	89.2%	9.0%	1.4%

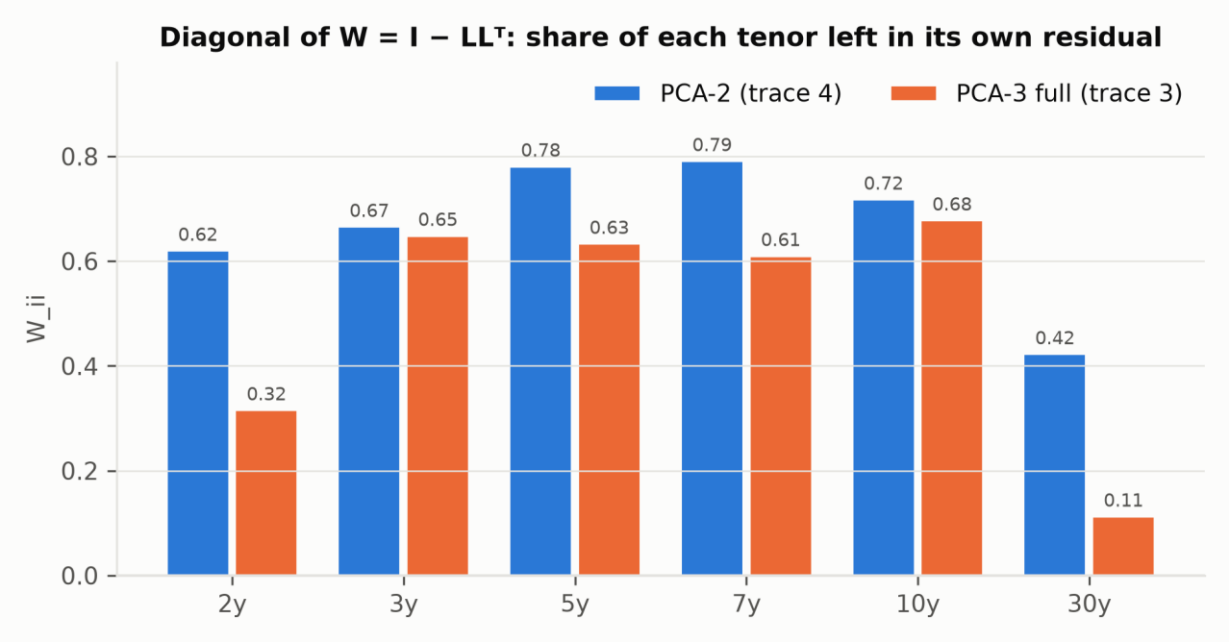


- Level moves every tenor the same way; slope is front vs back; curvature is belly vs wings.
- Refit every fold on that fold's own 5-year training window.

Fold with test window from 2022-12-12; eigenvector signs are arbitrary (level is negative on every tenor). Source: deck_numbers.md §B.

THE HEDGE MATRIX

$W = I - LL^T$ keeps what the factors miss



$$W = I - L\,L^T, \quad L^T W = 0$$

$$\text{trace}(W) = 6 - 2 = 4$$

5y package, one row of W (bp of yield)

2y	3y	5y	7y	10y	30y
−0.23	−0.24	+0.78	−0.19	−0.14	−0.06

- $W_{ii} = \|W e_i\|^2$: the share of tenor i's own direction left after the hedge.
- Butterfly: long the 5y, short both sides, so no level or slope exposure.
- Hedging curvature too (PCA-3) leaves only 0.32 of the 2y and 0.11 of the 30y.

Fold trained 2018-01 to 2022-11. Package weights in bp of yield per unit of the 5y residual. Source: deck_numbers.md §B.

THE SIGNAL · OU S-SCORE

Fade the residual when it is stretched

```
C_t = a + b·C_{t−1} + e          last 60 bars, refit every bar
m = a/(1 − b)      σ_eq = sd(e)/√(1 − b²)      z = (C_t − m)/σ_eq
trade only if 0 < b < 1 and half-life = ln 2/κ = −ln 2/ln b < 30 bars
```

PCA-2 tenor	Half-life, bars	σ_eq, bp	Screen pass
2y	6.36	1.60	88.0%
3y	5.50	0.91	94.6%
5y	5.98	1.47	89.6%
7y	7.08	1.11	87.1%
10y	6.28	1.10	91.6%
30y	6.99	1.53	88.7%

- Entry ∈ {1.0 … 2.5}, exit ∈ {0 … 0.75}, set per fold by training net Sharpe; stop at |z| > 4.
- z > 0: tenor yield high vs the factors → buy the tenor, sell the hedge legs.
- 5y on 2023-03-13: z = +2.87, σ_eq = 2.80 bp → 8.03 bp from equilibrium.
- Half-life is used only in the screen, never for sizing or timing.

PCA-2, live test bars passing the screen. Code: sleeves/pca.py:50-69, models/ou.py:44-90. Source: deck_numbers.md §A, §C, §K.

ONE TRADE, END TO END

The 5y after SVB, entry to exit

Entry (signal bar)	2023-03-13
z at entry	+2.87
σ_eq at entry	2.80 bp
Gap from equilibrium, z × σ_eq	8.03 bp
Size (vol 1.90 bp/day)	−0.526 risk units
Exit	2023-04-10, 20 bars, z = 0.49
Residual move	−6.20 bp (in our favour)
Gross / cost / net, risk units	+3.26 / 0.52 / +2.74
Gross / cost / net, $1m	+$15.3k / $2.4k / +$12.9k

Leg	Weight	$ per bp
2y	−0.233	+600
3y	−0.241	+622
5y	+0.781	−2,012
7y	−0.185	+477
10y	−0.145	+372
30y	−0.057	+147
Net / gross		+206 / 4,230

- Long 5y DV01 against short wings: the trade is the residual, not duration.

$/bp = P&L on $1m for +1 bp in that yield. Member pca2_ust_5y, fold from 2022-12-12, bands 2.5 / 0.5. Source: deck_numbers.md §K.

HIT RATIO · CONDITIONAL GAIN AND LOSS

Wins are frequent; losses are twice as big

Model	Hit gross	Hit net	Gross	Cost	Net	W	L	W ÷ L
Headline	68.1%	54.5%	+0.71	1.33	−0.62	1.94	3.68	0.53
PCA-3 sparse	83.1%	57.6%	+1.74	2.08	−0.33	1.52	2.86	0.53
PCA-2 full	67.2%	54.4%	+0.69	1.39	−0.70	2.04	3.97	0.51
Pairwise NN	69.0%	54.0%	+0.48	1.10	−0.62	1.89	3.56	0.53
Fly 2s5s10s	56.7%	51.7%	−0.19	1.07	−1.26	1.90	4.63	0.41
Fly 3s5s7s	74.4%	44.2%	+1.12	2.27	−1.15	1.74	3.43	0.51

Breakeven hit rate

`1/(1 + W÷L) = 1/1.53 = 65%`

Net hit rates are 44–58%: below it for every model.

Expectancy

`p·W − (1 − p)·L`

L ≈ 2×W everywhere, so only a stop or a regime gate shrinks it.

Why the band is wide

`round trip 0.73 bp, fixed`

Captured ≈ (entry − exit)·σ_eq, so the band widens: entry at the grid top (2.5) in 45–70% of PCA-2 folds.

Units: risk units = one trailing daily σ at entry (not σ_eq, not bp). PCA-3 sparse: distinct trades. Source: deck_numbers.md §D.

QUIET VS VOLATILE ENTRIES

Quiet markets can't pay a fixed cost

Model	Tercile cuts, bp/day	Round trip ÷ vol, low	Net, low	Net, mid	Net, high
Headline	0.50 / 1.03	1.78	−1.25	−0.27	−0.33
PCA-3 sparse	0.92 / 1.54	2.32	−0.48	+0.04	+0.48
PCA-2 full	0.47 / 0.77	1.96	−1.47	−0.20	−0.45
Pairwise NN	1.23 / 3.01	1.30	−1.77	−0.01	−0.09

- Entry vol = the trailing EW-60 residual vol on the entry bar (bp/day), the same vol used for sizing.
- Position = 1/vol, so a quiet market buys more DV01 per unit of edge: the round trip costs 1.3–2.3× a day's move in the low tercile.
- Low-vol trades lose for every model; only PCA-3 sparse's high-vol third earns (+0.48σ per trade).

Basis: member trades; terciles pooled per model over the full sample (ex post, descriptive). Net in risk units. Source: deck_numbers.md §F.

PER DOLLAR

$1 million in: earned vs paid

$1.07

cost per $1 of gross P&L, PCA-3 sparse

$2.10

cost per $1 of gross P&L, headline

Model	Gross $/yr	Cost $/yr	Net $/yr	Take-home per $1	Gross DV01 $/bp	Net DV01 $/bp
Headline	45,500	95,800	−50,300	−$1.10	6,632	446
PCA-3 sparse	100,500	107,600	−7,100	−$0.07	9,374	175
PCA-2 full	38,500	96,700	−58,200	−$1.51	7,596	498
Pairwise NN	35,600	55,500	−20,000	−$0.56	3,856	263

- Take-home per $1 of gross = 1 − cost ÷ gross.
- Gross DV01 $3.9k–9.4k per bp, net only $175–498: long and short the curve at once.

Basis: $1m of capital at the 10% vol target (not notional); additive returns. DV01 is the mean over live days. Source: deck_numbers.md §G.

MODEL COMPARISON

Best and worst models side by side

Model	Gross SR	Net SR	Breakeven ×	IC, 10d	Trades/yr	Cost, % pa
PCA-3 sparse (best)	1.21	−0.09	0.93	0.33	7.9	10.8
Headline (PCA-2 + credit)	0.55	−0.60	0.48	0.17	13.1	9.6
Pairwise NN	0.46	−0.26	0.64	0.13	7.7	5.6
PCA-2 full	0.45	−0.67	0.40	0.20	10.6	9.7
Fly 3s5s7s	0.38	−0.77	0.32	—	43 total	10.1
Fly 2s5s10s (worst)	0.11	−0.42	0.22	—	60 total	3.9

- Every model is positive gross and negative net.
- The fixed flies, traded with the same rules, earn far less gross (0.11–0.38): the PCA hedge adds signal.

Trades/yr = trades ÷ (live bars ÷ 252), identical trades merged; headline 13.1 = 426 trades. Source: deck_numbers.md §E.

COST ROUND 1

Freezing size helps; nothing clears zero

Net Sharpe at cost ×	0.5×	1×	2×	Cost % pa
Headline · baseline	−0.03	−0.60	−1.64	9.6
Headline · freeze size	+0.10	−0.35	−1.20	8.2
Headline · freeze + gate	+0.06	−0.22	−0.76	4.5
PCA-3 sparse · baseline	+0.58	−0.09	−1.28	10.8
PCA-3 sparse · freeze size	+0.63	+0.07	−0.95	9.5
PCA-3 sparse · freeze + gate	+0.33	+0.09	−0.38	3.9

- Freeze size (no parameter): cost −11 to −14%, net SR +0.16 to +0.25.
- Vol gate (X fitted per fold, a trial): halves cost again, and halves PCA-3 sparse gross (1.17 → 0.56).
- Headline still negative at 1x (−0.22).
- PCA-3 sparse +0.09 at 1x; deflated Sharpe 0.002 at 47 trials.

Bands and the gate level X are fitted at 1x; 0.5x and 2x re-price the same positions. Source: deck_numbers.md §H, §J.

PROJECT EVOLUTION

Seven steps, 47 trials, one answer

1	2	3	4	5	6	7
Pairwise, 10 spreads	**Universe to 27–31**	**4 design revisions**	**PCA models**	**Filters + curve rules**	**Step-8 experiments**	**Cost round 1**
NN gate 0.51 / −0.01	NN gate 0.46 / −0.26	no Sharpe saved	PCA-3 sparse 1.21 / −0.09	16 variants	best 1.20 / −0.00	best 0.56 / +0.09

47

documented trials: 11 models, 8 cost filters, 8 curve rules, 4 revisions, 12 experiments, 4 cost-round configs

0.002

deflated Sharpe, net, of the best variant (PCA-3 sparse freeze + vol gate)

0.42

deflated Sharpe, gross, same variant: a 42% probability the true gross Sharpe is above zero

No git history: steps from dated table snapshots. Pairs are gross / net Sharpe. Source: deck_numbers.md §J.

WARNINGS

Read the best result with care

1 · Hedge defect, 2010–13

`+68.2 / −120.9 / +68.9`

3y hedged by a near-singular 2y/5y/10y solve. Trades in the four folds lost 34 risk units across 13 distinct trades (20 from the 3y).

2 · Overlap

`6 members = 2.3 effective bets`

The 2y, 3y and 5y members are often one portfolio. Counting each distinct portfolio once: net SR −0.097 vs −0.087.

3 · Selection

`47 trials → net DSR 0.002`

Several experiment settings were chosen after full-sample results were seen.

4 · Optimistic signal

`half-life 6.4 vs 103 bars`

The 60-bar AR(1) is biased fast (small sample); the entry band sits at the grid top in 45–70% of PCA-2 folds.

The defect is deliberately not fixed: a fix chosen after seeing results is another trial. Source: deck_numbers.md Defect, §I, §J, §A, §C.

STEP 1 · COVARIANCE AND PCA

Find the moves the whole curve makes together

$$\Sigma = \mathrm{cov}(\Delta y,\ \text{training window})$$
$$\Sigma = V\,\Lambda\,V^{\mathsf{T}}$$
$$L = \text{first } k \text{ columns of } V$$
$$\text{explained} = (\lambda_1 + \dots + \lambda_k) \,/\, \Sigma\lambda$$

- **Changes, not levels:** daily changes in bp are close to stationary
- **Covariance, not correlation:** demeaned, not standardized, so higher-vol tenors weigh more
- **Training window only:** full-sample PCA would be look-ahead

Loadings, fold 0 (trained 1990–1994)

Tenor	Level	Slope	Curve
2y	-0.504	-0.718	-0.456
5y	-0.551	-0.126	+0.632
10y	-0.509	+0.353	+0.252
30y	-0.429	+0.587	-0.574

Two factors explain **97.9%** of daily variance here, three explain 99.4%. Across all 33 folds the two-factor median is 98.2%.

Code: sleeves/pca.py, factor_loadings, changes.cov()

STEP 2 · THE HEDGE

Strip out level and slope, trade what is left

$$r_t = W \Delta y_t,\ W = I - L L^{\top}\ (k = 2) \Rightarrow L^{\top} W = 0\text{: no level or slope exposure}$$

One row of W is one tradeable package. The 5y package, fold starting Dec 2022:

2y	3y	5y, own tenor	7y	10y	30y
-0.23	-0.24	+0.78	-0.19	-0.14	-0.06

Own weight, 2y, fold 0

$1 - (0.504^2 + 0.718^2)$

= 0.230

Trace of W, fold 0

$n - k = 4 - 2$

= 2

Variance removed, 5y

$1 - (0.90 / 6.09)^2$

= 97.8%

Curvature is kept

$W L_3 = L_3$

27% of gross DV01

Package Σ|w| = 1.64, so one-way cost = 0.25 × 1.64 = 0.41 bp

STEP 3 · THE SIGNAL

Fade the residual when it is stretched

```
X_t = Σ r_s (cumulative residual)
X_t = a + b·X_t-1 + e (last 60 bars)
half-life = ln 2 / (-ln b)
m = a / (1 - b)
σ_eq = sd(e) / √(1 - b²)
z = (X_t - m) / σ_eq
```

- Tradeable only if 0 < b < 1 and half-life < 30 bars
- Enter at |z| above the band (fitted per fold, 1.0–2.5); exit inside 0–0.75; stop at |z| > 4

WORKED EXAMPLE · 5Y, MAR 13, 2023

Input	Value
z at entry (band 2.5)	+2.87
Half-life, 60-bar fit	4.7 bars
σ_eq	2.80 bp
Forecast: (z – exit) × σ_eq	6.6 bp
Round-trip cost	0.82 bp
Result, net	+2.74σ

z > 0 means the bond is cheap: buy the 5y, sell the hedge legs in package proportions.

Avellaneda–Lee (2010) s-score and speed filter; settings not tuned

STEP 4 · SIZING AND DV01

Every trade carries the same risk, not the same DV01

```
p_t = signal_t / vol_t
vol = EWMA sd of r, span 60,
floored at 25% of its median
cost_t = |p_t - p_t-1| × Σ|w| × 0.25 bp
```

- **DV01:** the dollar change for a 1 bp move in yield
- **Risk unit (σ):** one trailing daily sd of the member
- Members equal-weighted; sleeve scaled to 10% annual vol

$6,632

gross DV01 per bp, headline, on $1m

$446

net DV01 per bp: close to rate-neutral

SAME 5Y PACKAGE, TWO MARKETS

Mar 2023: vol 1.90 bp/day → 0.53 units
Mar 2024: vol 0.43 bp/day → 2.34 units

4.4× the DV01, and 4.4× the cost, for the same risk

A 1 bp move in one leg moves the headline 0.07–0.15 bp of capital ($700–1,500 per $1m)

RESULTS BY MODEL

Four models, one pipeline: all positive gross, all negative net

Model	What it hedges	Gross SR	Net SR	Breakeven	Eff. bets	Trades
Headline	Level, slope; plus 3 credit pairs	0.55	−0.60	0.48×	8.7	426
PCA-3 sparse	Level, slope, curve; 4 legs	1.21	−0.09	0.93×	2.3	255
PCA-2 full	Level, slope; every tenor	0.45	−0.67	0.40×	5.7	344
Pairwise NN	17 hand-picked spreads, Kalman	0.46	−0.26	0.64×	14.4	252
2s5s10s fly	Fixed weights (benchmark)	0.11	−0.42	0.22×	1	60

Gross rises with how much curve structure is hedged: the fixed fly 0.11, PCA-2 0.45, PCA-3 sparse 1.21.

Net is set by cost. Only PCA-3 sparse comes close, and it is the least diversified.

Headline = PCA-2 full curve plus HY/IG OAS, CDX cash basis, HYG vs CDX. Breakeven = gross ÷ cost

RESULTS BY TRADE

Hit ratio and conditional gain and loss

Model, per trade (σ)	Hit, gross	Hit, net	Gross	Cost	Net	Win ÷ loss
Headline	68%	54%	+0.71	1.33	−0.62	0.53
PCA-3 sparse	83%	58%	+1.74	2.08	−0.33	0.53
PCA-2 full	67%	54%	+0.69	1.39	−0.70	0.51
Pairwise NN	69%	54%	+0.48	1.10	−0.62	0.53

Breakeven hit rate

`1 / (1 + 0.53) = 65%`

Gross hit rates clear it. Net (54–58%) do not.

Conditional gain and loss

`W ≈ +1.9σ  L ≈ -3.6σ`

Headline, implied: many small wins, a few large losses.

Expectancy

`p·W - (1 - p)·L`

0.54 × 1.9 − 0.46 × 3.6 = −0.62σ a trade

σ = one trailing daily sd of the member. W and L implied from net mean, hit rate and win ÷ loss

WHY IT LOSES · 1 OF 3

Cost is fixed in bp. The edge scales with volatility.

PCA-2 5y residual	Mar 2023, stressed	Mar 2024, quiet
Residual vol at entry	1.90 bp/day	0.43 bp/day
One-way package cost	0.41 bp	0.42 bp
Round trip ÷ vol	0.43	1.95
Gross	+3.26σ	+3.47σ
Cost	0.52σ	2.18σ
Net	+2.74σ	+1.29σ
Cost share of gross	16%	63%

THE MECHANIC

Both trades were right and earned about the same gross. The quiet one held 4.4× the DV01, so it paid 4.4× the cost per unit of risk.

```
hole at entry ≈ 2 × 0.41 bp ÷ vol
at 0.4 bp/day ≈ 2σ
```

Source: s3_trades_pca2; cost = 0.25 bp × Σ|w| per unit, one way

WHY IT LOSES · 2 OF 3

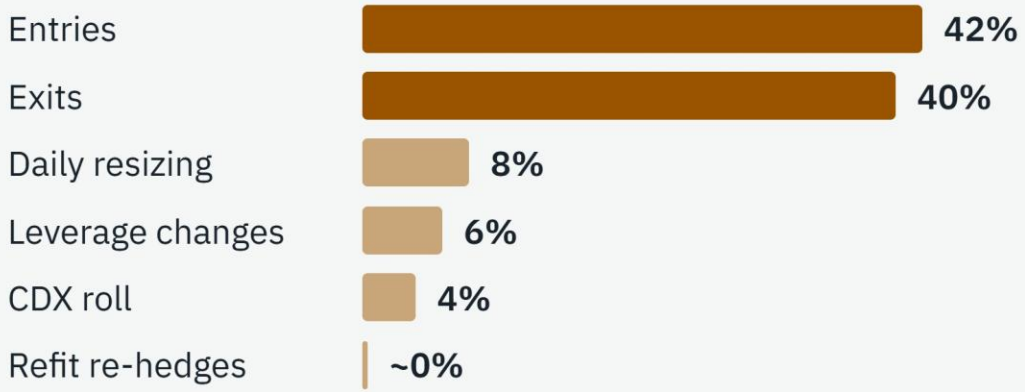
Quiet-market trades lose, and round trips are the bill

Headline trades by residual vol at entry (terciles)

Entry vol	bp/day	Trip ÷ vol	Net/trade
Low	0.39	1.78	−1.25σ
Mid	0.68	1.01	−0.27σ
High	4.40	0.59	−0.33σ

PCA-3 sparse makes **+0.49σ** a trade in its high-vol third and loses **−0.48σ** in its low-vol third. Gross per trade is similar across terciles; cost is what changes.

Where headline cost goes (9.6% of capital a year)



82% is opening and closing trades. Netting legs across members saves only 0.5–5.5%.

Diagnostics Steps 1 and 4; entries plus exits are 75–87% of cost across models

WHY IT LOSES · 3 OF 3

The 60-bar OU fit promises more than arrives

103
bars: half-life on the 5-year training window

6.4
bars: median half-life the 60-bar signal uses

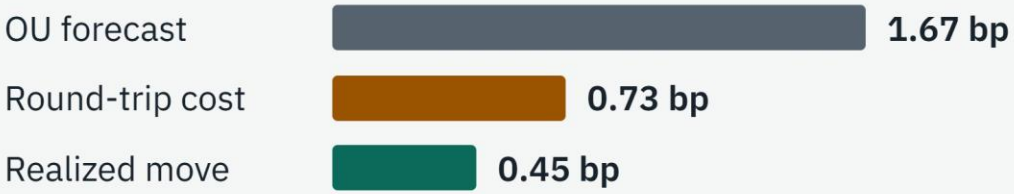
3.5
bars: half-life on entry days

10
bars: median holding, PCA-2

Most of the gap is small-sample bias (Kendall)

```
b = 2^(-1/103) = 0.9933
bias ≈ (1 + 3b) / 60 = 0.066
b^ ≈ 0.927 → half-life ≈ 9.1 bars
simulated: 8.3 before screen, 7.6 after
```

So the forecast overstates the reward (PCA-2)



A “trade only if forecast beats cost” rule keeps 92% of entries.

Forecast = median at entry; realized = mean per trade; halving entries needs k ≈ 2.4

WHERE THE LOSSES SIT

Trades still open after 20 bars lose the money

Model (risk units)	Held 20+	Share	Their net	All others
Headline	67	16%	−167	−96
PCA-3 sparse	61	24%	−172	+87
PCA-2 full	58	17%	−135	−107
Pairwise NN	60	24%	−150	−7

A TIME STOP DOES NOT FIX IT

These trades lose in the first 20 bars (headline −65, sparse −75), then win some back (+31, +47).

Tested: a 20-bar stop cuts PCA-3 sparse's net Sharpe by 0.26.

Hypothesis, not tested: the 60-bar mean is refit every bar, so when a residual trends the mean follows it. z never reaches the 4σ stop and the trade closes "at the mean" at a loss.

Source: trades.md, holding-period table; OU half-lives on training windows are 3–7 bars at entry

PCA-3 SPARSE, READ WITH CARE

The best variant comes with three warnings

1 · A BAD HEDGE, 2010–13

Before the 7y existed, the 3y was hedged with 2y, 5y and 10y, which span almost the same factors.

```
+68 / -121 / +69
Σ|w| = 259 → 65 bp a unit
```

Trades in those folds: **−34** risk units (13 distinct trades).

2 · FEWER BETS THAN MEMBERS

Four tenors and three factors leave one possible package. The 2y, 3y and 5y all use {2, 3, 5, 7}; the 10y and 30y share {5, 7, 10, 30}.

```
6 members
= 2.3 effective bets
```

3 · PICKED AFTER THE FACT

It is the best of 47 documented trials. The deflated Sharpe asks whether that best result beats luck.

```
DSR net 0.000
DSR gross 1.00
```

The gross edge survives. The net result does not.

Worst fold condition number 155 passed the cond < 1e8 guard; a cap near 20, or on Σ|w|, would catch it

WHAT WE TESTED

Trading less helps, but nothing turns net positive

Experiment, walk-forward	Δ net SR: P2 / H / P3s	Verdict
Resize only on a 25% change	+0.07 / +0.06 / +0.09	Helps all; PCA-3s reaches −0.00
Entry band 2.5	+0.33 / +0.20 / +0.05	Only 2.7–6 trades a year
Entry band 3.0	+0.48 / +0.33 / +0.01	0.6–3 trades a year: too few
MOVE top-tercile gate	+0.14 / +0.09 / +0.05	Small help
Leg netting across members	+0.09 / +0.07 / +0.01	Saves 0.5–5.5% of cost
20-bar time stop	+0.02 / +0.01 / −0.26	Hurts the best model
Leave-one-out residual	−0.06 / −0.08 / −0.68	Worse everywhere
Leung–Li optimal levels	turnover 5× lower	PCA-2 gross Sharpe falls to −0.05

Most parameters were chosen after seeing results: treat these gains as upper bounds.

Bands re-fitted per fold under each rule; pipeline reproduces production bit for bit with no change applied

A CONTRAST: DIRECTIONAL SLOPE TRADES

Trading the slope on purpose survives costs

2s10s rule, same engine	Gross SR	Net SR	Max DD
Policy rule (2y ±25 bp)	0.68	0.45	−36%
Slope momentum	0.48	0.09	−73%
Always flattener	0.11	0.05	−53%
Always steepener	−0.11	−0.18	−105%
PCA headline	0.55	−0.60	−179%
PCA-3 sparse	1.21	−0.09	−54%

THE DIFFERENCE IS COST

Entering a slope trade costs **0.17σ**; a PCA-2 residual costs **0.53σ**.

Curve rules lose 1–4% a year to cost; PCA sleeves lose 10–11%.

Correlation with the PCA headline: −0.07 to +0.01.

Policy rule: flattener while hikes are priced (+9.6% a year), steepener while cuts are priced (+11.6%). It is a directional macro trade, so it belongs with the trend sleeve, not here.

From 1995; DV01-neutral 2s10s; drawdowns additive; policy rule untuned but only a few dozen independent calls

LIMITATIONS

What the P&L does not include

No carry or financing

Traded in cash-yield bp. Any carry the book collects would show up as alpha.

Yields, not instruments

On-the-run yields switch issue at auctions. Real trades are bonds or futures.

Cost is an assumption

Wider spreads before 2008 take net SR to −1.12 (headline) and −0.63 (sparse).

Additive P&L

Summed, not compounded, so drawdowns can exceed 100%. A real book would have stopped.

Credit is a proxy

HY and IG OAS are index spreads and cannot be traded as modeled.

Multiple testing

47 documented trials; most experiment settings were chosen after seeing results.

Leakage tests pass; the remaining look-ahead risk is in researcher choices, which the report flags

NEXT STEPS

Fewer, better round trips, then a clean test

1	**Freeze position size at entry**	Mechanical; removes most resize and leverage cost
2	**Cost-aware entry, k fitted per fold**	Targets the quiet-market third that loses
3	**Cap the sparse hedge weights**	Correctness fix for 2010–13; counts as a trial
4	**Replace the 60-bar OU forecast**	Bias-correct it, or learn reversion size in bp
5	**Price futures as a cost scenario**	About 0.1 bp a leg is an estimate to confirm
6	**Add carry; keep an untouched holdout**	Required before any net conclusion

Stop rule: if PCA-3 sparse still fails net on untouched data after steps 1–4, drop it.

Each change declared before it runs, tested walk-forward and added to the deflated-Sharpe trial count

TAKEAWAY

The curve does mean-revert after the hedge. At 0.25 bp a leg, the reversion is smaller than the round trip.

Signal

Real: IC 0.17–0.33, gross deflated Sharpe 1.00

Cost

82% round trips, worst in quiet markets

Path

Drop the expensive trades, then a clean holdout

Questions

APPENDIX A · KEY TERMS, STATISTICS

Variance, covariance and the signal

Term	Meaning	Formula	In this project
Variance	Spread of a series around its mean	$E[(x - \mu)^2]$	Raw 5y: $6.09^2 \approx 37$ bp²
Standard deviation	Typical daily move	√variance	5y resid. 0.90 vs 6.09 raw
Covariance	How two series move together	$E[(x - \mu x)(y - \mu y)]$	PCA input, bp², train only
Correlation	Covariance scaled to −1 … +1	cov / (σx · σy)	Not used for the PCA
Variance explained	Share of variance in the top k PCs	$\Sigma\lambda_1 \ldots \lambda_k / \Sigma\lambda$	98.2% median, k = 2
Variance removed	Share a hedge takes out	$1 - (\sigma_res / \sigma_raw)^2$	97.8% for the 5y
Half-life	Time to close half the gap	ln 2 / (−ln b)	103 vs 6.4 bars
z-score (s-score)	Distance from mean in sd units	(X − m) / σ_eq	Enter at |z| of 1.0–2.5
Information coeff.	Rank corr. of signal and next move	Spearman(−z, Δ10)	0.17 headline, 0.33 sparse

bp² = basis points squared; λ = eigenvalue of the covariance matrix

DV01, hit ratio and conditional P&L

Term	Meaning	Formula	In this project
DV01	$ change for a 1 bp yield move	Δ value per 1 bp	$6,632 gross per $1m
Risk unit (σ)	One trailing daily sd of a member	p = signal / vol	1σ = one typical day
Hit ratio	Share of trades that make money	wins / trades	68% gross, 54% net
Conditional gain	Average win, given a win	E[P&L | P&L > 0]	≈ +1.9σ, headline
Conditional loss	Average loss, given a loss	E[P&L | P&L < 0]	≈ −3.6σ, headline
Expectancy	Average P&L per trade	p·W − (1 − p)·L	−0.62σ, headline
Sharpe ratio	Annual return per unit of vol	μ·252 / (σ·√252)	0.55 gross, −0.60 net
Cost drag	Sharpe lost to cost	C / σ, both annual	0.096 / 0.084 = 1.14
Breakeven multiple	Cost level where net P&L is zero	gross / cost	0.48× headline
Deflated Sharpe	P(true SR > 0) after N trials	Bailey–López de Prado	0.000 net, N = 47
Effective bets	Independent bets in the book	$(\Sigma\lambda)^2 / \Sigma\lambda^2$	2.3 of 6, PCA-3 sparse

W, L = average win and loss; conditional gain and loss implied from net mean, hit rate and win ÷ loss

APPENDIX C · LIKELY QUESTIONS

Short answers to the hard questions

Is this arbitrage?

No, statistical relative value. No-arbitrage binds only up to trading cost; these dislocations sit inside that band.

Why does annualized vol matter?

Sharpe is return over vol, and cost drag is cost over vol: 9.6% ÷ 8.4% removes 1.14 Sharpe.

Where is the machine learning?

PCA, refit each fold, and an MLP gate (16, 8) on the credit pairs that predicts 10-bar reversion.

The IC is positive. Why does it lose?

IC gives direction, not size. The mean realized move is 0.45 bp against a 0.73 bp round trip.

Did you overfit?

Walk-forward, purge, embargo, headline named in advance. With 47 trials, net DSR is 0.000; gross is 1.00.

Why not just trade less?

Leung–Li cut turnover 5× and gross Sharpe fell to −0.05. Drop the expensive trades, keep the good ones.

References

Avellaneda, M., and Lee, J.-H. (2010). Statistical arbitrage in the U.S. equities market. Quantitative Finance, 10(7), 761–782.

Bailey, D. H., and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. Journal of Portfolio Management, 40(5), 94–107.

Bloomberg L.P. (2026). Generic US Treasury yields (USGG2YR–USGG30YR), LF98OAS and LUACOAS OAS indices, CDX HY 5Y, HYG. Bloomberg Terminal.

Board of Governors of the Federal Reserve System (2026). DFEDTAR, DFEDTARU. FRED, Federal Reserve Bank of St. Louis.

Engle, R. F., and Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. Econometrica, 55(2), 251–276.

Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. Econometrica, 59(6), 1551–1580.

Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. Journal of Basic Engineering, 82(1), 35–45.

Kendall, M. G. (1954). Note on bias in the estimation of autocorrelation. Biometrika, 41(3/4), 403–404.

Leung, T., and Li, X. (2015). Optimal mean reversion trading with transaction costs and stop-loss exit. International Journal of Theoretical and Applied Finance, 18(3), 1550020.

Litterman, R., and Scheinkman, J. (1991). Common factors affecting bond returns. Journal of Fixed Income, 1(1), 54–61.